AFRICAST

TIDY TIME SERIES & FORECASTING IN R

Cross-validation and residual diagnostics



Session 9 of 10

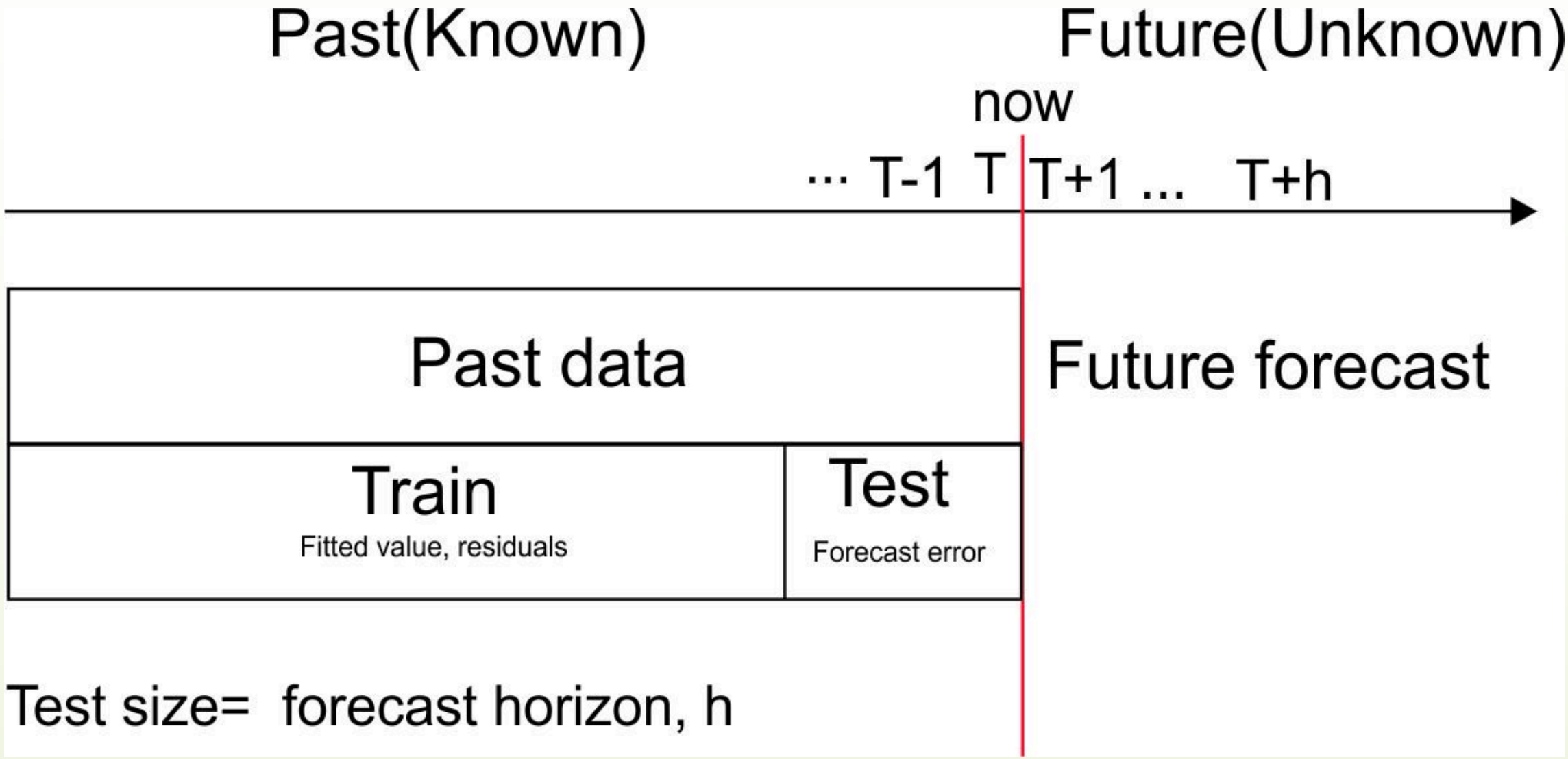
SECTION 1 OF 3

Time series cross-validation

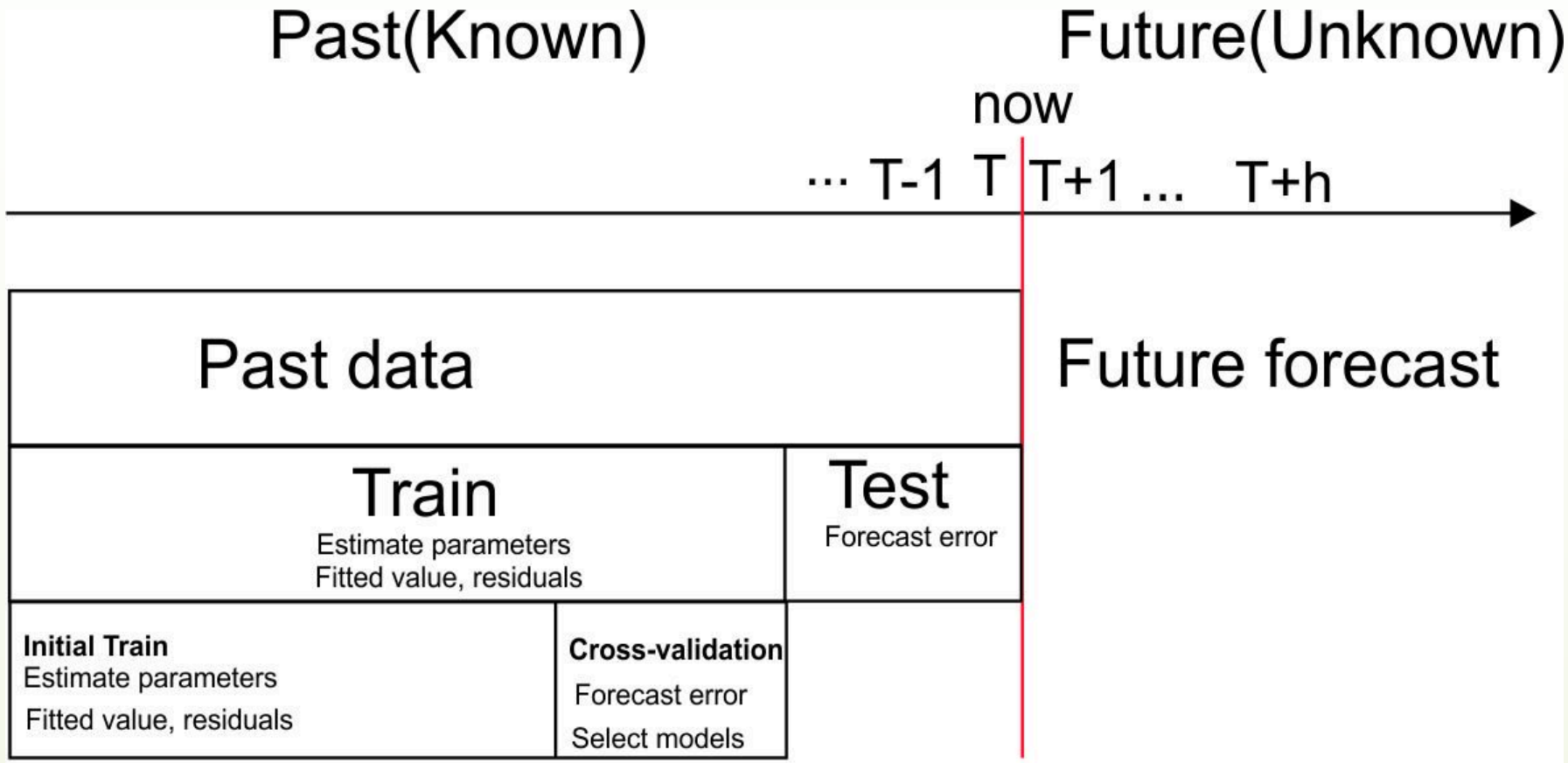
1 **Time series cross-validation** 2 Residual diagnostics 3 Recap



Issue with traditional train/test split



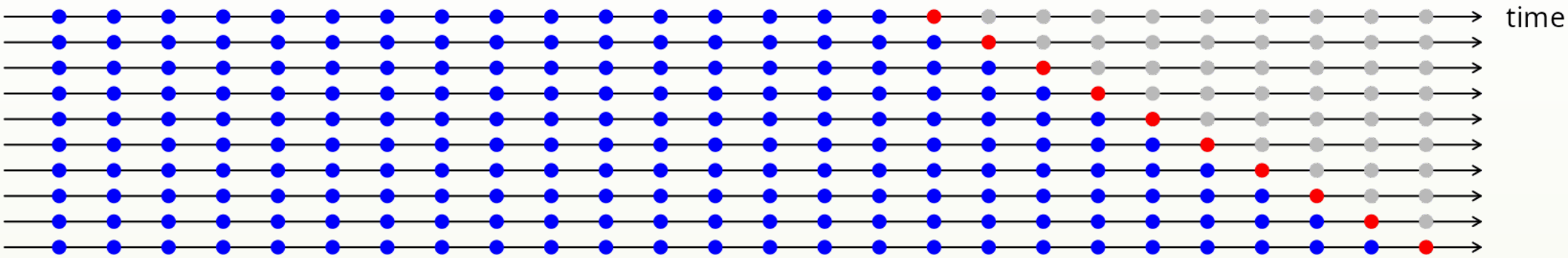
Forecasting from a rolling origin



Test size= forecast horizon, h

Cross-validation size=nb of experiment+h-1

Time series cross-validation



- Forecast accuracy averaged over test sets.
- Also known as “evaluation on a rolling forecasting origin”

Creating the rolling training sets

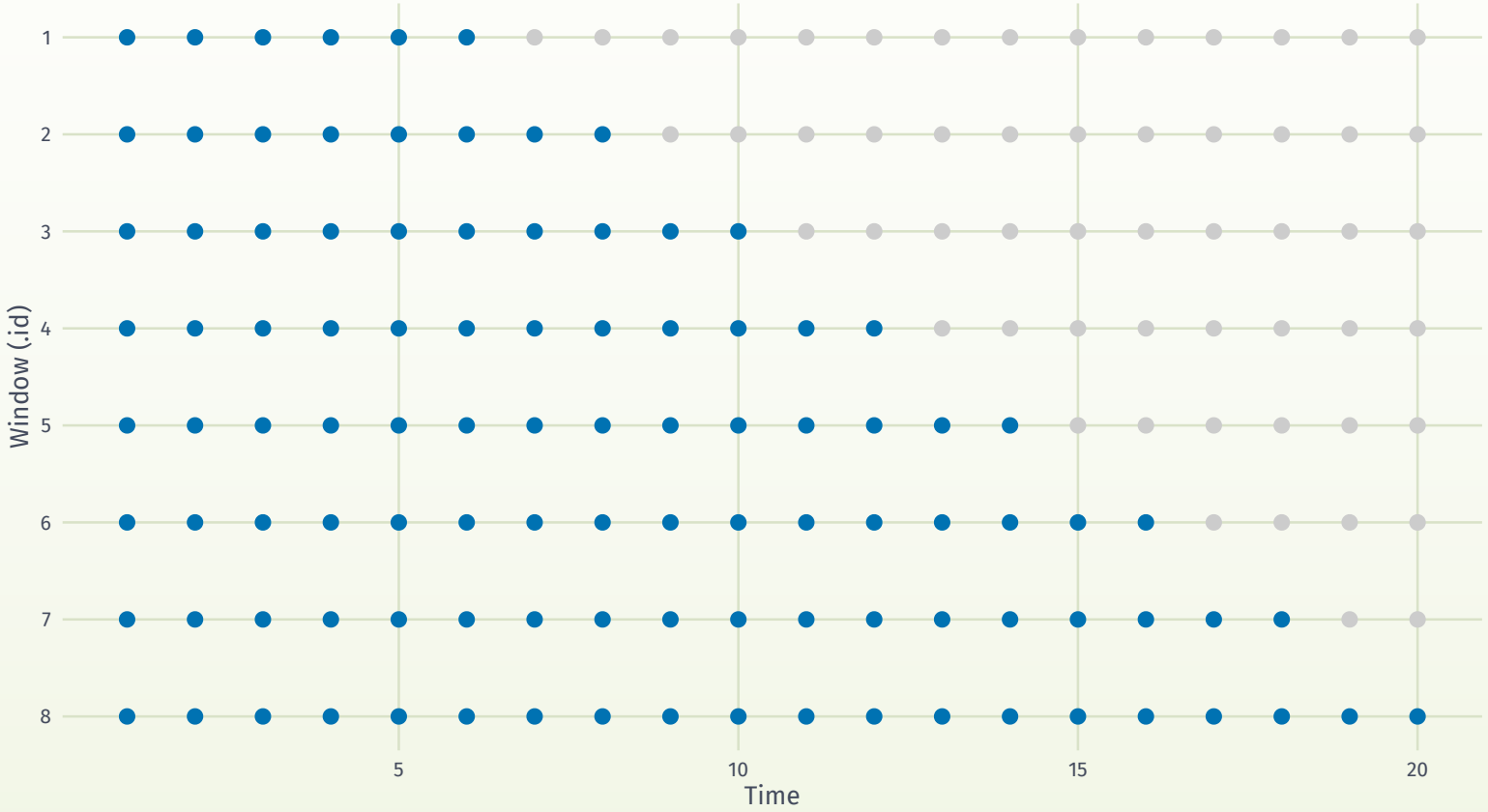
Three functions roll a tsibble into windows, numbered by a new `.id` column.

- `stretch_tsibble()`: extends a growing window with new data.
- `slide_tsibble()`: shifts a fixed length window through the data.
- `tile_tsibble()`: moves a fixed length window without overlap.

For time series cross-validation, stretching windows are most commonly used.

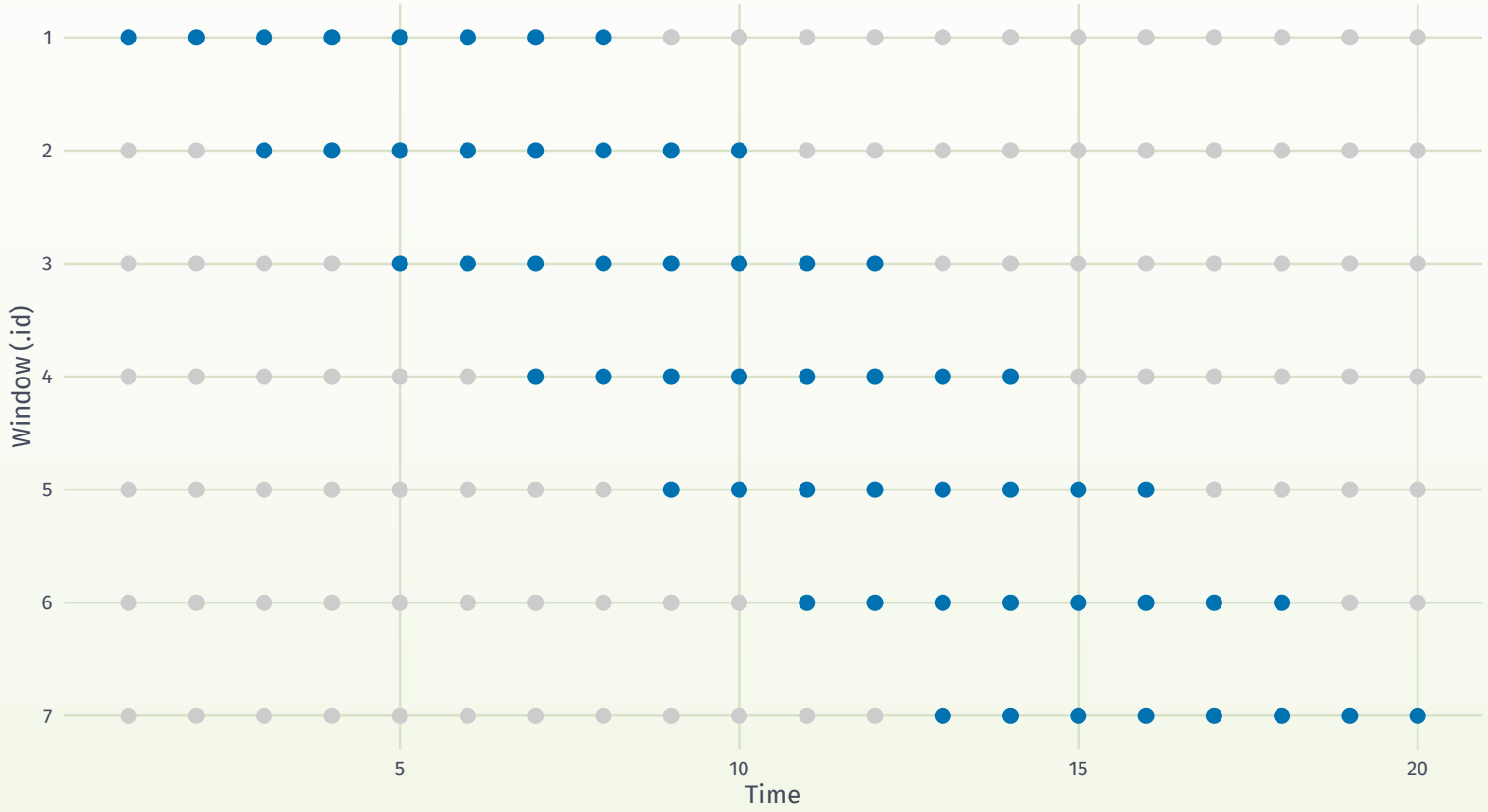
Stretch: `stretch_tsibble()`

```
x <- tsibble(t = 1:20, index = t)
stretch <- x |> stretch_tsibble(.init = 6, .step = 2)
```



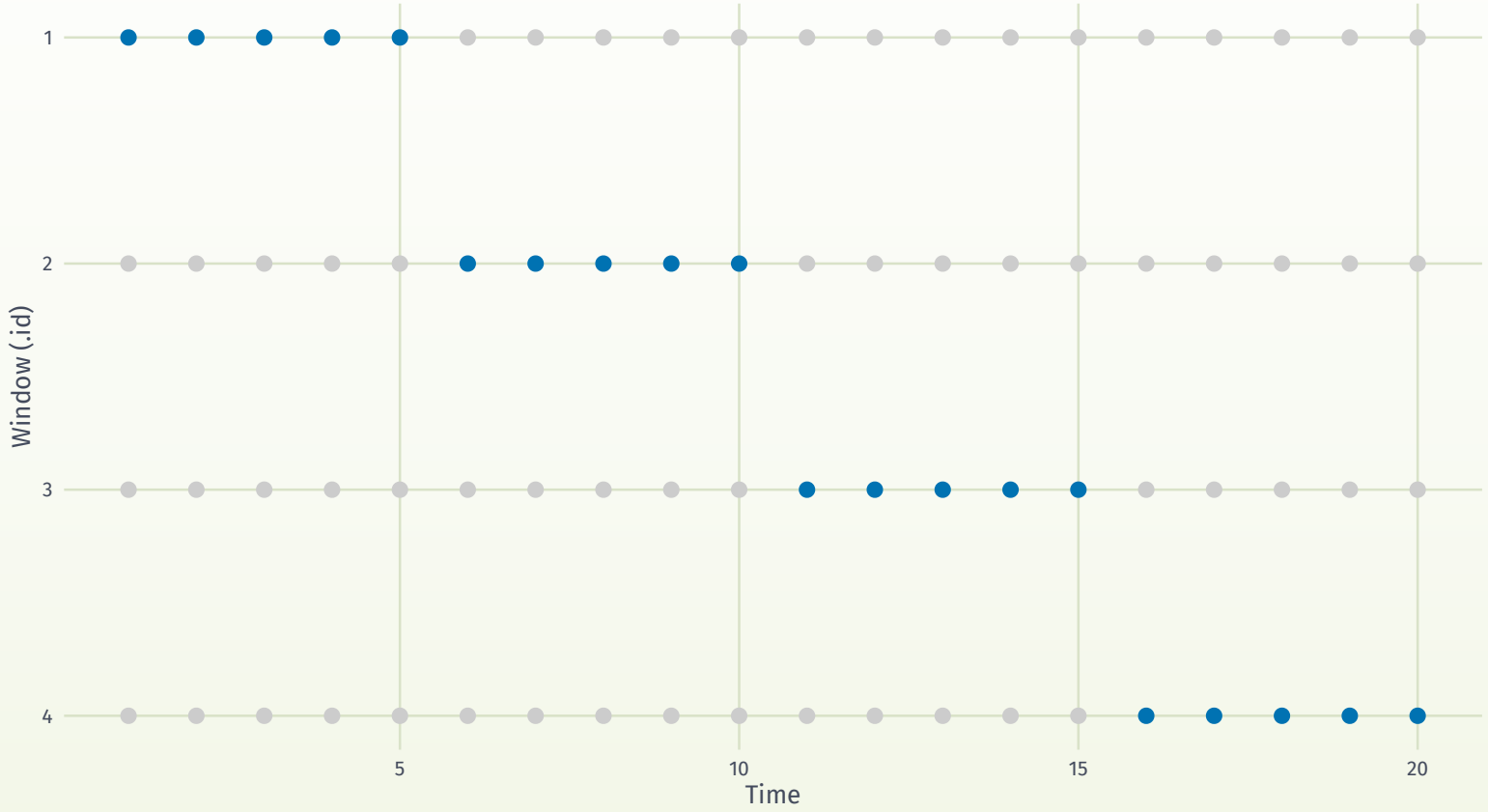
Slide: `slide_tsibble()`

```
slide <- x |> slide_tsibble(.size = 8, .step = 2)
```



Tile: `tile_tsibble()`

```
tile <- x |> tile_tsibble(.size = 5)
```



Rolling origins for drug sales

Monthly antidiabetic drug sales in Australia ($ millions), from 2000. Stretch with a minimum length of 24, growing by 1 each step.

```
forecast_horizon <- 12
test <- antidiabetic_drug_sale |>
  slice((n()-forecast_horizon+1):n())
train <- antidiabetic_drug_sale |>
  slice(1:(n()-forecast_horizon))
drug_sale_tcsv <- train |>
  stretch_tsibble(.init = 24, .step = 1)
```

```
# A tsibble: 3,819 x 3 [1M]
# Key:       .id [67]
     Month  Cost   .id
     <mth> <dbl> <int>
1 2000 Jan 12.5      1
2 2000 Feb  7.46     1
3 2000 Mar  8.59     1
4 2000 Apr  8.47     1
# i 3,815 more rows
```

Fit models to each window

Estimate seasonal naïve, naïve and drift models for each window.

```
drug_fit_tr <- drug_sale_tcsv |>
  model(
    snaive = SNAIVE(Cost),
    naive = NAIVE(Cost),
    drift = RW(Cost ~ drift())
  )
```

```
# A mable: 67 x 4
# Key:     .id [67]
    .id   snaive   naive         drift
  <int>  <model> <model>       <model>
1     1 <SNAIVE> <NAIVE> <RW w/ drift>
2     2 <SNAIVE> <NAIVE> <RW w/ drift>
3     3 <SNAIVE> <NAIVE> <RW w/ drift>
4     4 <SNAIVE> <NAIVE> <RW w/ drift>
# i 63 more rows
```

Forecasts from each window

Produce 12-step-ahead forecasts from each model and window.

```
drug_fc_tr <- drug_fit_tr |>
  forecast(h = forecast_horizon)
```

```
# A fable: 2,412 x 5 [1M]
# Key:     .id, .model [201]
    .id .model    Month        Cost .mean
  <int> <chr>     <mth>      <dist> <dbl>
1     1 snaive 2002 Jan  N(14, 1.7) 14.5
2     1 snaive 2002 Feb   N(8, 1.7)  8.05
3     1 snaive 2002 Mar  N(10, 1.7) 10.3
4     1 snaive 2002 Apr N(9.8, 1.7)  9.75
# i 2,408 more rows
```

Which method will win?

Which method do you expect to forecast antidiabetic drug sales best?

Seasonal naïve	0	0%
Naïve	0	0%
Drift	0	0%



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africast-cv-residuals

0 responses

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Cross-validated accuracy

```
drug_fc_tr |>
  accuracy(
    antidiabetic_drug_sale,
    measures = list(
      point_accuracy_measures,
      interval_accuracy_measures,
      distribution_accuracy_measures
    )
  ) |>
  select(.model, RMSE, MAE, MASE, winkler, CRPS)
```

```
# A tibble: 3 × 6
  .model  RMSE   MAE  MASE winkler  CRPS
  <chr>  <dbl> <dbl> <dbl>   <dbl> <dbl>
1 drift   3.95  3.05  1.89    27.0  2.43
2 naive   3.87  2.96  1.84    25.0  2.31
3 snaive  2.42  1.94  1.20    15.3  1.42
```

Errors are averaged over all origins and all 12 horizons.

From cross-validation to forecasts

Seasonal naïve wins the cross-validation. How do you produce the forecasts you deliver?

Use the forecasts from the last window	0	0%
Refit seasonal naïve on all the data	0	0%
Average the forecasts from every window	0	0%



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0 responses

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SECTION 2 OF 3

Residual diagnostics

1 Time series cross-validation
2 **Residual diagnostics**
3 Recap



Forecasting residuals

Residuals in forecasting: difference between observed value and its fitted value (one-step forecast) on the training data: $e_t = y_t - \hat{y}_{t|t-1}$.

Assumptions

1. $\{e_t\}$ uncorrelated. If they aren't, then information left in residuals that should be used in computing forecasts.
2. $\{e_t\}$ have mean zero. If they don't, then forecasts are biased.

Useful properties (for prediction intervals)

3. $\{e_t\}$ have constant variance.
4. $\{e_t\}$ are normally distributed.

Facebook closing stock price

```
fb_stock <- gafa_stock |>
  filter(Symbol == "FB")
fb_stock |>
  autoplot(Close) +
  labs(x = "Date", y = "Closing price (US$)")
```



Naïve model of Facebook's stock price

```
fb_stock <- fb_stock |>
  mutate(trading_day = row_number()) |>
  update_tsibble(index = trading_day, regular = TRUE)
fit <- fb_stock |> model(NAIVE(Close))
augment(fit)
```

```
# A tsibble: 1,258 x 7 [1]
# Key:       Symbol, .model [1]
   Symbol .model       trading_day Close .fitted .resid .innov
   <chr>  <chr>              <int> <dbl>   <dbl>  <dbl>  <dbl>
 1 FB     NAIVE(Close)           1  54.7    NA    NA      NA
 2 FB     NAIVE(Close)           2  54.6    54.7 -0.150 -0.150
 3 FB     NAIVE(Close)           3  57.2    54.6  2.64   2.64
 4 FB     NAIVE(Close)           4  57.9    57.2  0.720  0.720
 5 FB     NAIVE(Close)           5  58.2    57.9  0.310  0.310
 6 FB     NAIVE(Close)           6  57.2    58.2 -1.01  -1.01
 7 FB     NAIVE(Close)           7  57.9    57.2  0.720  0.720
 8 FB     NAIVE(Close)           8  55.9    57.9 -2.03  -2.03
 9 FB     NAIVE(Close)           9  57.7    55.9  1.83   1.83
10 FB     NAIVE(Close)          10  57.6    57.7 -0.140 -0.140
# i 1,248 more rows
```

Fitted values

```
augment(fit) |>
  ggplot(aes(x = trading_day)) +
  geom_line(aes(y = Close, colour = "Data")) +
  geom_line(aes(y = .fitted, colour = "Fitted")) +
  labs(x = "Trading day", y = "Closing price (US$)", colour = NULL)
```



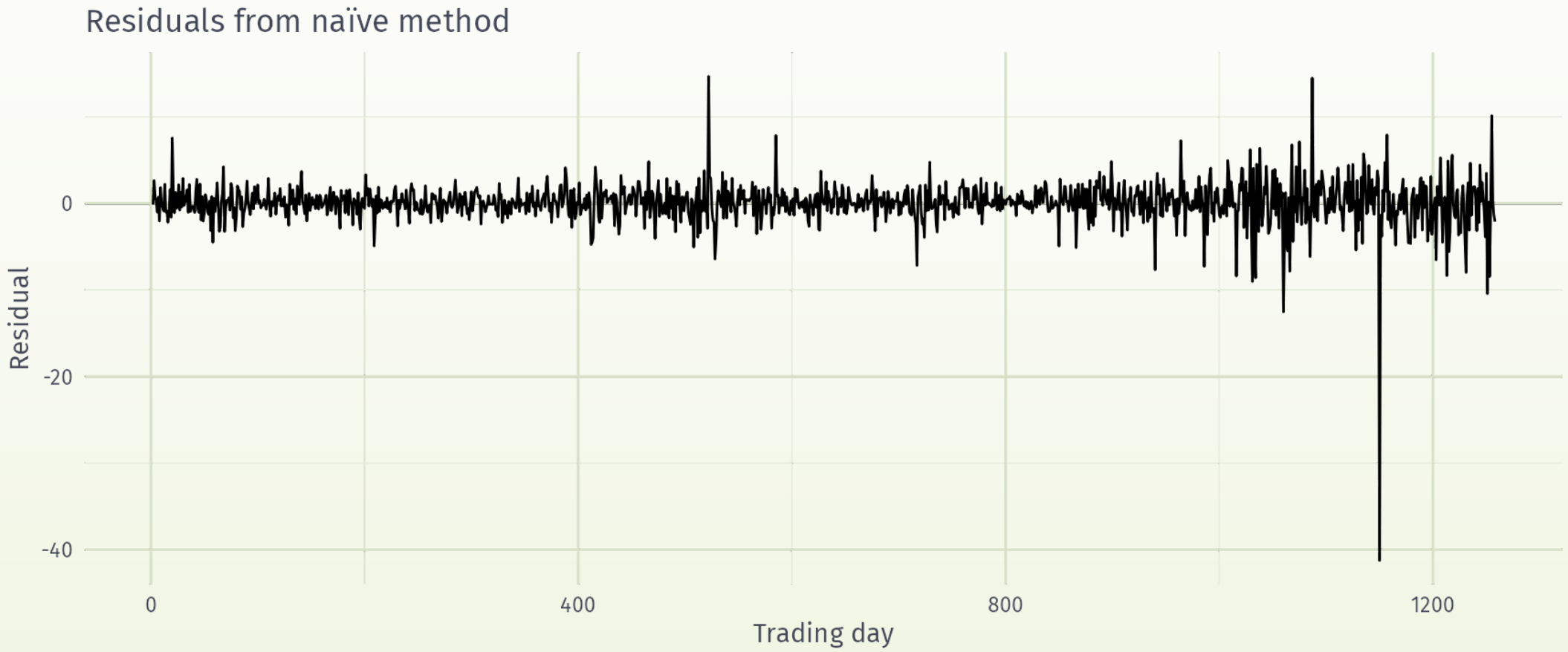
Fitted values (last 158 trading days)

```
augment(fit) |>
  filter(trading_day > 1100) |>
  ggplot(aes(x = trading_day)) +
  geom_line(aes(y = Close, colour = "Data")) +
  geom_line(aes(y = .fitted, colour = "Fitted")) +
  labs(x = "Trading day", y = "Closing price (US$)", colour = NULL)
```



Residuals over time

```
augment(fit) |>
  autoplot(.resid) +
  labs(x = "Trading day", y = "Residual", title = "Residuals from naïve method")
```



Constant variance?



Do these residuals have constant variance?

Yes **0** 0%

No **0** 0%

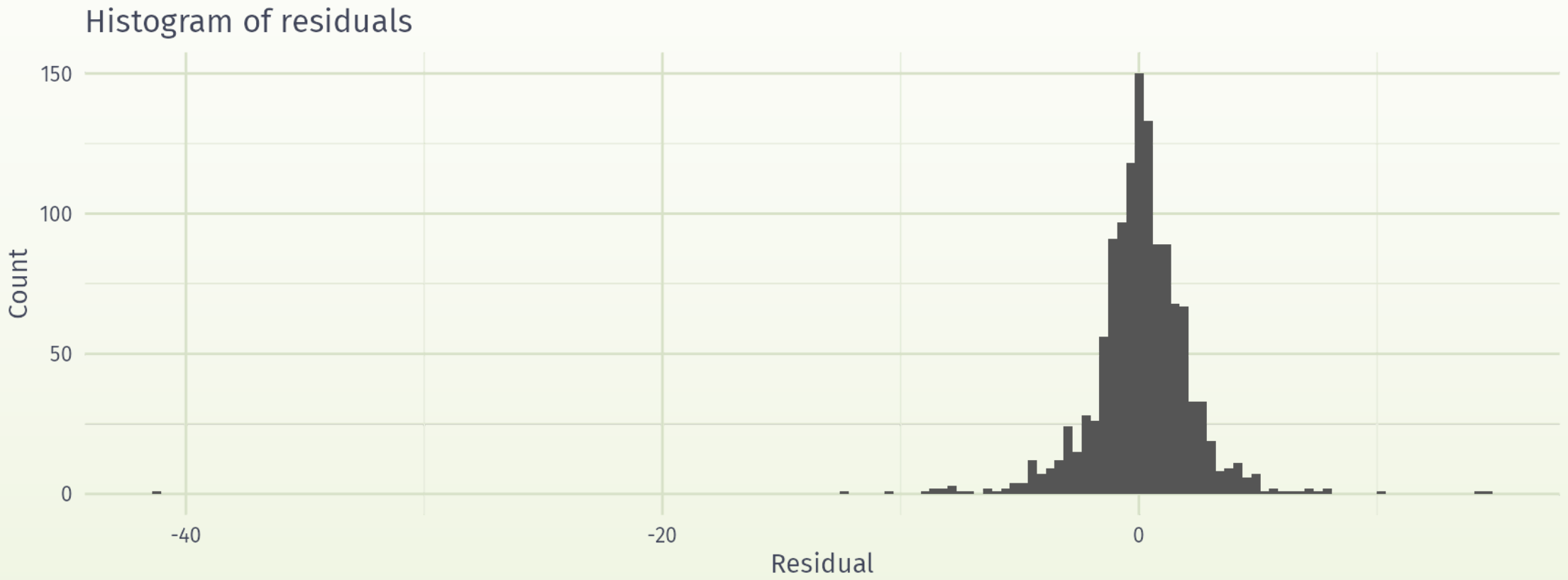
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0 responses

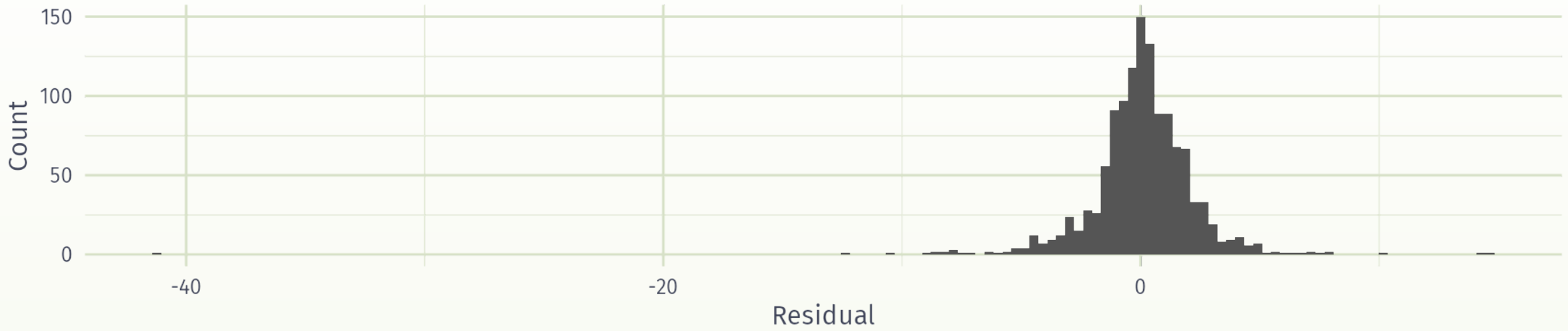
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Histogram of residuals

```
augment(fit) |>
  ggplot(aes(x = .resid)) +
  geom_histogram(bins = 150) +
  labs(x = "Residual", y = "Count", title = "Histogram of residuals")
```



Normally distributed?



Do these residuals look normally distributed?

Yes **0** 0%

No **0** 0%

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0 responses

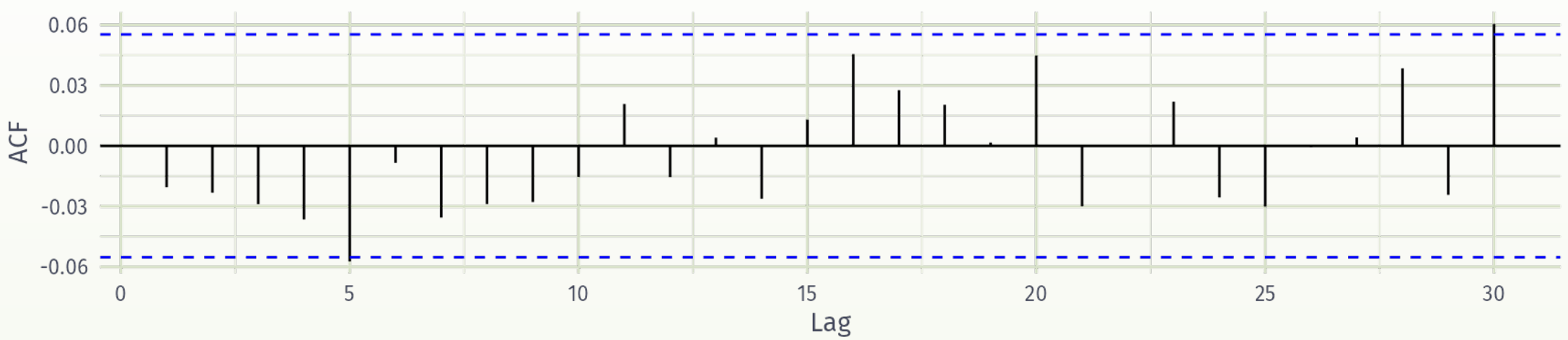
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ACF of residuals

```
augment(fit) |>
  ACF(.resid) |>
  autoplot() +
  labs(x = "Lag", y = "ACF", title = "ACF of residuals")
```



White noise?



Do these residuals look like white noise?

Yes **0** 0%

No **0** 0%

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0 responses

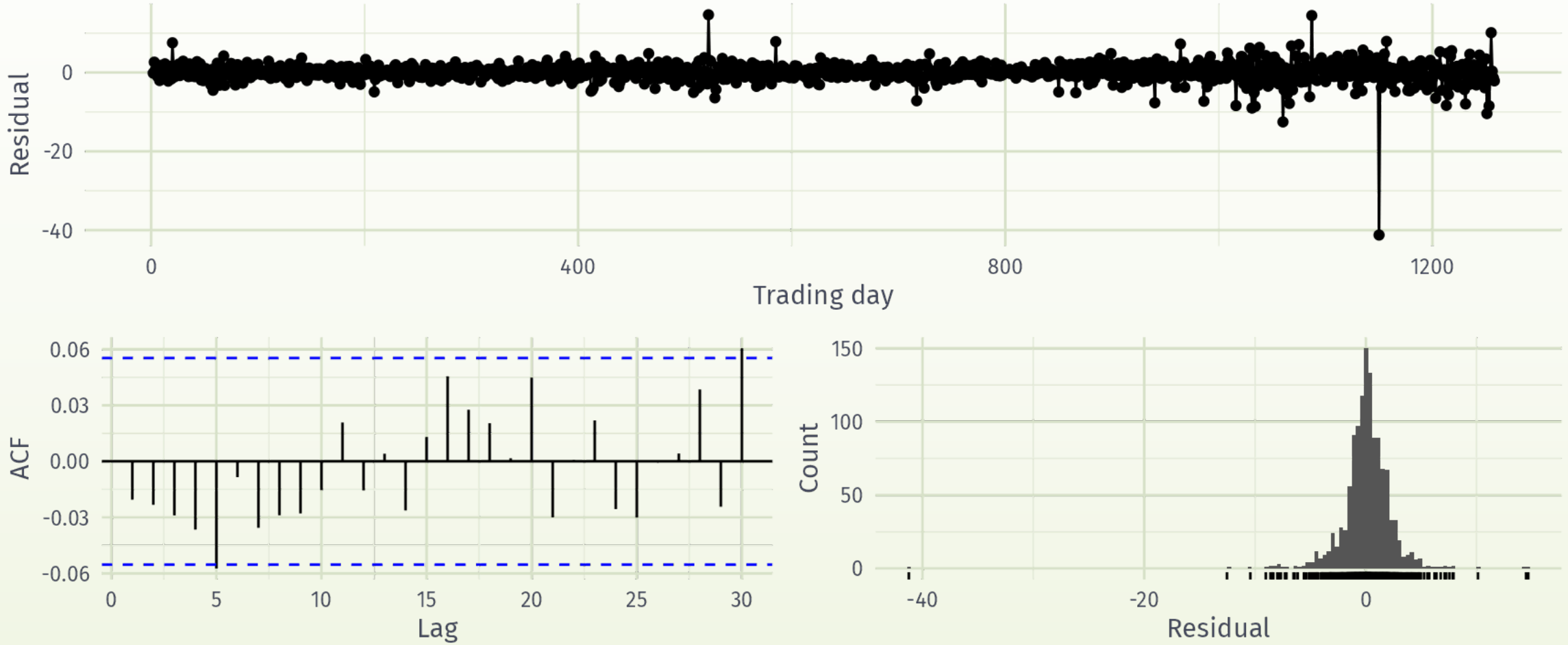
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Checking the ACF of residuals

- We assume that the residuals are white noise (uncorrelated, mean zero, constant variance). If they aren't, then there is information left in the residuals that should be used in computing forecasts.
- So a standard residual diagnostic is to check the ACF of the residuals of a forecasting method.
- We *expect* these to look like white noise.

Combined diagnostic graph

```
fit |> gg_tsresiduals()
```



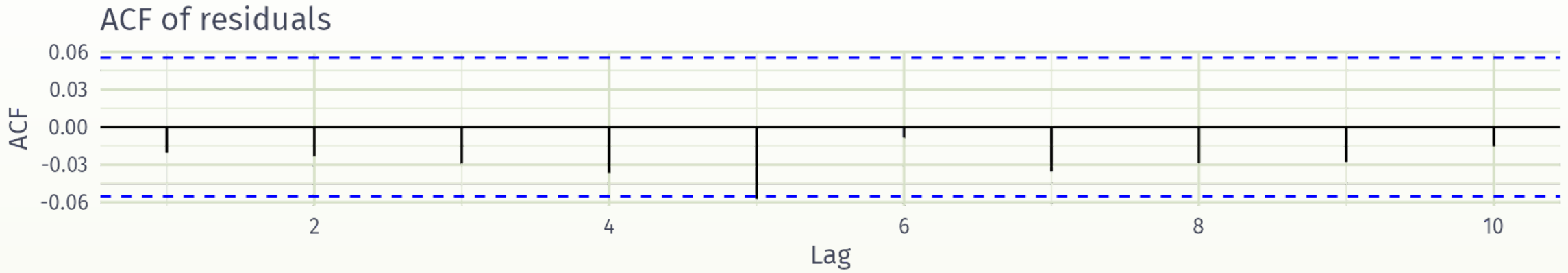
Ljung-Box test

Test whether *whole set* of r_k values is significantly different from zero set.

$$Q = T(T+2)\sum_{k=1}^{\ell}(T-k)^{-1}r_k^2 \qquad \text{where } \ell = \text{max lag and } T = \text{\# observations}$$

- If each r_k close to zero, Q will be **small**.
- If some r_k values large ($+$ or $-$), Q will be **large**.
- Suggested lags: $\ell = 10$ for non-seasonal data, $\ell = 2m$ for seasonal data (but at most $T/5$).
- If data are WN and T large, $Q \sim \chi^2$ with ℓ degrees of freedom.
- For residuals of a model with K estimated parameters, use $\ell - K$ degrees of freedom (`dof` argument).

Ljung-Box test of the naïve residuals




```
augment(fit) |> features(.resid, ljung_box, lag = 10)
```

```
# A tibble: 1 × 4
  Symbol .model       lb_stat lb_pvalue
  <chr>  <chr>          <dbl>     <dbl>
1 FB     NAIVE(Close)    12.1     0.276
```

$p = 0.276 > 0.05$: no evidence against white noise.

What the residuals tell us

- Mean close to zero: the forecasts are unbiased.
- Uncorrelated: no information is left in the residuals.
- Variance is not constant: it grows later in the series.
- Heavy tails and a large outlier: the residuals are not normal.

The point forecasts are fine, but the prediction intervals may be inaccurate.

SECTION 3 OF 3

Recap

1 Time series cross-validation
2 Residual diagnostics
3 **Recap**



Recap

- **Time series cross-validation** averages forecast accuracy over many rolling origins, not a single test set.
- Roll the training data into windows with `stretch_tsibble()`, fit models to each window, forecast and compare with `accuracy()`.
- **Residual diagnostics** check that the best model's residuals are uncorrelated with mean zero, and ideally have constant variance and are normal.
- Check them with `gg_tsresiduals()` and the Ljung-Box test before forecasting.